APP1: Be Amazed

Algorithmic Problem Solving

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September 9, 2025

Source: Jason Hawkes in *Britain from Above*

Learning objectives:
Using divide and conquer algorithms and recursive data structures to devise algorithms.

I. What is APP?

In APP, learning is based on the collective resolution of a "real" problem, with the goal being learning precise knowledge. There are three levels of pedagogical objectives:

1st level — Direct discipline objectives:

- Analysing a problem before any coding/experiment;
- Proposing pertinent tools that would help solving the problem (data structures, algorithms...);
- Predicting the behavior of your algorithms (quality of solution, complexity...);
- Analysing results with a critical eye.

2nd level — Transverse objectives:

- Working in cooperation with a team;
- Knowing how to look for information;
- Developing autonomy, motivation, and initiative;
- Reinforcing the capacity to make choices and to evaluate risks;
- Managing time and deadlines.

3rd level — Auto-reflexion: being capable of analysing, understanding, and evaluating one's own learning process. Being capable of continually improving by learning from one's experience.

I.1. Method

In APP, work is divided in multiple phases, some of them in groups during class sessions, some of them individually at home.

1st session — "go"

- Understand the task at hand;
- Identify what you already know;
- Formulate leads to follow.

In-between — Realize the actions planned (individual work).

2nd session — "return" (and following sessions)

- Share work of everyone (compare and evaluate);
- Build a solution;
- Self-review.

Make sure that, at any time, the following tasks are fulfilled:

- everyone is encouraged to participate;
- responsibilities are assigned;
- the group is organized.

I.2. Team Work

To learn how to work in a team, we want you to form a cooperative learning group based along the following values:

- Solidarity: "All for One & One for All!";
- Listening and mutual respect;
- Respect of work requirements and schedules;
- Good atmosphere;
- Implication of everyone (no stowaway!).

The objective is to optimize the *active participation* of every student inside the group. Each and everyone of the group members need to pay attention to these rules and that everyone is implicated. In order for the team work to be efficient, we advise you to attribute specific roles that will ease the well-functioning of the group (see next page).

II. Team Work Needs Organization!

In order for your group to really function as a team, we strongly advise you to distribute in your group the following three specific roles. The other team members still have to take individual notes, participate in discussion, and review resource materials.

Chair

- Makes sure the group follows the steps and the timings;
- Makes sure the discussion is dully noted by the recorder;
- Regularly informs the group of the remaining time;
- Animates the discussion:
  - shares the talk time between people, encourages participation or temperates interventions;
  - encourages the group to clarify developed ideas;
  - synthesizes if required.

Scribe

- Is at the blackboard or the flip-chart if there is any;
- Listens carefully to the group discussions;
- Translates the essential of what he hears on the board (support and memory of group discussion);
- Does *not* filter information;
- Organizes the board following the steps (do not erase anything).

Recorder

- Keeps a trace (written and complete) of the group production; (**this will be used for the report!**)
- Transmits this trace to every group member.

For this APP, inscribe the name of the volunteers for each of those roles (do not take a role you have already been!):

Chair _____

Scribe _____

Recorder _____

Tutor

He guides the group: help them not to get lost, and encourages to explore directions...
He knows the answer to the problem, but it is *your job*, students, to do the work. So don't be surprised if he sometimes refuses to answer directly your questions. This will for instance be the case if he considers this question has not yet been fully debated in the group.

III. APP1: The Maze

III.1. APP Objectives

After this APP1, you will be capable of:

- Identifying when a problem can be solved using divide and conquer;
- Using binary trees;
- Performing various kind of walk-through on binary trees;
- Analysing the complexity of an algorithm.

III.2. Session Organization

This APP will last four sessions. Personal work is expected between sessions ("trap", for *TRAvail Personnel* in french).

Session 1. Opening (1h15 min.) Discovery of the problem by personal reading then group discussions.

Trap 1. ⇒ Reading provided material on data structures. Write (on paper) a data structure for the problem, and the algorithm to create the maze.

Session 2. (1h30) Sharing work in group about the data structure and algorithm to generate the maze. Decide collectively on the best way to do it. Start thinking about the size of your data structure (individually then in group).

Trap 2. ⇒ Reading of the rest of the notes: walk-through of data structures. Design (on paper) an algorithm to solve the maze.

Important: Prepare at least two–three questions to ask the tutor on the reading material for next session.

Session 3. (1h30) Reply to student's questions; sharing work on the algorithm to solve the maze and group work on the algorithm.

Trap 3. ⇒ Reading notes on algorithmic complexity. Analysis of the data structure and algorithms on the maze.

Session 4. Closing (1h30 min.) Group discussion on complexity and writing of the deliverable (must be given to the tutors before the end of the session).

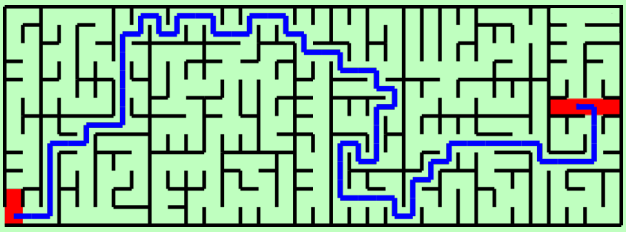
Important: Between each and every session, personal work is expected.

Note: You are not required to do an actual implementation of the algorithms. However, if you feel that it would help you understand and solve the problem, we give a skeleton in C at Moodle - APS - Maze Code, including in particular drawing functions for the SVG image format.

Situation — problem

Minos is not happy with the new maze Deadalus has invented. He comes to your team asking for a way to generate mazes randomly so that no one can have access to the blueprints. The maze will be built under the palace, and should have only one exit. But there are many entries possible through trap-doors in the palace.

Knowing the man's propensity for using inventions on their creators, you decide accept the king's request, but secretly swear to find a way to keep track of any generated maze, as well as find a way to navigate to the exit, just in case...



You realize you can build a simple maze by just dividing the allocated space using a full wall separating one of the two dimensions (e.g., the longest dimension), and placing only one door at an arbitrary position on this wall, creating now two spaces, then applying the same reasoning on those new spaces.

For this APP, you are asked to solve two problems:

1. Given the width and height of the initial area, generate a maze;
2. Find the (unique, hence shortest) path from any position in the maze to the exit.

These two problems will first need to be solved in an algorithmic way (i.e., on paper), then their complexities must be analysed.

III.3. Resources

This APP builds on knowledge that should already be integrated at the Master level. It requires recursive data structures, such as lists and trees, and a good understanding of algorithm complexity.

It is required that you read the notes given on the moodle (see below). If you are unfamiliar with those notions, it is imperative to read also the chapters of the book reference.

- Moodle - APS - A Recursive Data Structure: Trees

- Moodle - APS - Cost Analysis of Algorithms
- "Introduction to Algorithms" by Cormen, Leiserson, Rivest, and Stein (3rd edition);
  - Chapters 10 (Elementary Data Structures);
  - Chapter 12 (Binary Search Trees);
  - Chapter 17 (Red-Black Trees).
  - Chapters 1,2,3 (Introduction up to Growth of Functions);
  - Chapter 7 (Quick-sort);
  - Chapter 17 (Amortized analysis).

IV. Session 1: Discovery and Analysis of the Problem (1h15 min.)

Timing	*What to do during the session*
1 5 min.	Organize your group: attribute roles (chair, scribe, recorder) for the project. People must volunteer but not take a role already taken in a previous APP.
2 20 min.	**Warning:** individual phase (no discussion!) First approach of the problem: read the problem with care. Try to find by yourself elements of solutions to the problem.
3 45 min.	Share in the group your comprehension of the problem and your first ideas. Write on the board (or a common sheet of paper) all your thinking, while keeping a special place where to put element that require precisions by the tutor. Try to organize your thinking around the following points: • Focus for now on the maze generation; • Imagine the data structure you will need.
4 5 min.	Make sure to understand what is expected for the next session (see below).

Total: 75 min.

V. TRAP 1: Maze Generation

We remind you that personal work for next session is **required** and *not an option*!

- Read with attention the first part of the notes on data structures (see Section III.3). If you feel you need it, also consult the chapters of the book mentioned.
- Elaborate individually an algorithm for the maze generation. Make sure you also clearly state which data structures you use and what information you store in them. Be careful to describe clearly each step of your reasoning so it can be easily explained (and understood) by the other members of your group.

Warning: Important: it is crucial that everyone in the group does its share of work, as otherwise no work can be done during the next session. You are responsible for **your own learning**, but the learning of the other members of your group **depends also on you**!

VI. Session 2: Work Session (1h30)

Timing	*What to do during the session*
1 15 min.	Share your production of algorithms and data structures with your group. **Important:** Everyone must present something! It is the chair's duty to make sure of that.
2 40 min.	From your sharing, decide collectively on the data structures to use and the algorithm for the maze generation. This must be as clear as possible and written down as it will be used **directly** in the deliverable.
3 15 min.	**Important:** Individual phase Start analysing the data structure you choose in terms of size.
4 10 min.	Quick group discussion on the size of your data structure. Make sure everyone understands what needs to be done for next session.

Total: 80 min.

VII. TRAP 2: Maze Solving

We remind you that personal work for next session is **required** and *not an option*!

- Read with attention the second part of the notes on data structures (traversals of trees, see Section III.3). If you feel you need it, also consult the chapters of the book mentioned.

- Elaborate an algorithm that finds the path in the maze. Make sure you clearly explain how it uses the data structure that holds the maze. Be careful to describe clearly each step of your reasoning so it can be easily explained (and understood) by the other members of your group.

VIII. Session 3: Work Session (1h30)

Timing	*What to do during the session*
1 20 min.	Share the algorithms for solving the maze with your group. **Important:** Everyone must present something! It is the chair's duty to make sure of that.
2 30 min.	From your sharing, decide collectively on the best algorithm to use to solve the maze. This must be as clear as possible as this will be used in the deliverable.
3 20 min.	Start analysing the complexities of your algorithms.
4 20 min.	Discussion on algorithmic complexity. Present to the tutor the results of your discussion

Total: 90 min.

IX. TRAP 3: Complexity Analysis (expected work: 2h)

- Read with attention the notes on algorithmic complexity. If you feel you need it, also consult the chapters of the book mentioned.
- Perform a complexity analysis of your algorithms (generation and solving) and data structures.

X. Session 4: Work Session outside of class (1h30)

Timing	*What to do during the session*
1 20 min.	Compare the analysis of everyone. **Important:** Everyone must present something! It is the chair's duty to make sure of that.
2 45 min.	Discussion on algorithmic complexity. Present to the tutor the results of your discussion, then write it down clearly so it can be included in the deliverable.
3 20 min.	Finish writing the deliverable and hand it to your tutor.
4 5 min.	Do an individual assessment of your learning by filling the chart at the end of this document. Those points will be testing during the evaluation at the beginning of the next session.

Total: 90 min.

XI. Closing Session

Timing	*What to do during the session*
1 20 min.	Individual evaluation to test learning on APP1.
2 20 min.	Group analysis. • **Individually:** think of what happened in the group during the APP, and what could be improved. • Compare your results with the others members and draw conclusions. • Share your analysis with the tutor.
3 50 min.	*Discovery of the next APP...*

Total: 90 min.

XII. Your Learnings

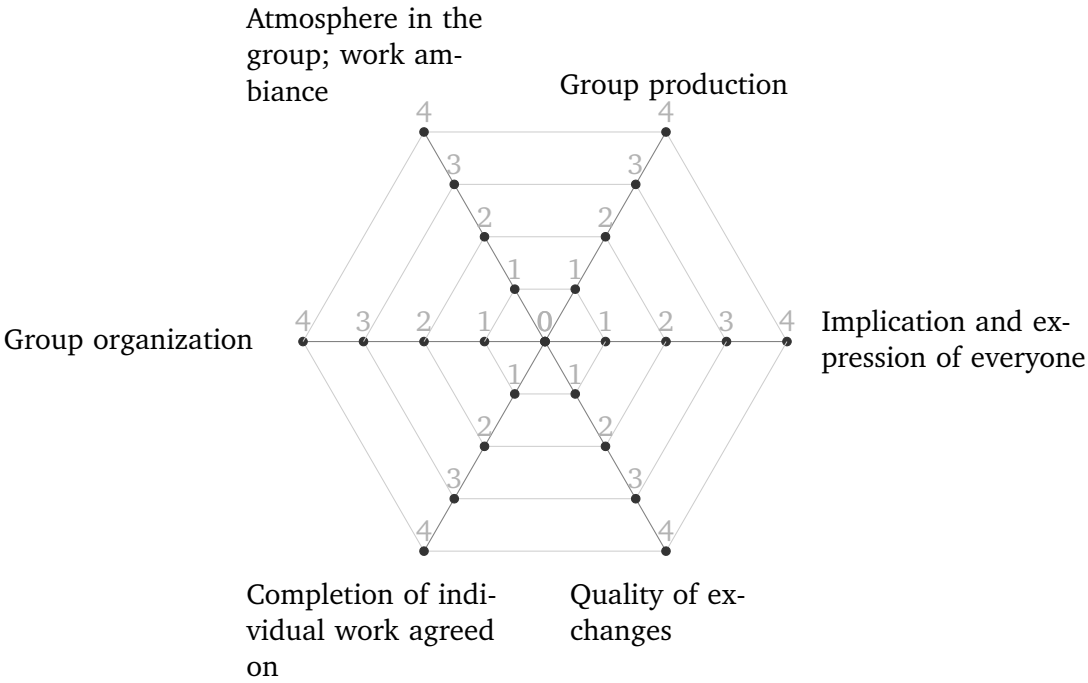
For each objective of this APP, evaluate by yourself your level of learning.

After this APP, you are capable of:	*Yes*	*No* (need to work on it)
1. Defining and using a binary tree in an algorithm and in an actual implementation;	_____	_____
2. Performing various basic operation on binary trees such as adding nodes, deleting them, searching them;	_____	_____
3. Explaining how the height of a binary tree relates to the number of nodes and which algorithms it impacts (in terms of complexity);	_____	_____
4. Using traversals on a binary tree;	_____	_____
5. Searching a path linking two nodes in a binary tree;	_____	_____
6. Stating the difference between cost and (asymptotic) complexity;	_____	_____
7. Analysing the complexity of a divide-and-conquer algorithm.	_____	_____

XIII. Auto-Evaluation of the Team Work — Circept

XIII.1. Assessment of group work

- **Individually:** fill the circept below and answer the questions on team work (10 min.)
- Compare your results with the others members and draw conclusions.
- Share your analysis with the tutor.



XIII.2. Give two positive points about your group work:

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XIII.3. Give two negative points about your group work:

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XIII.4. What commitment can you take to improve your group work?

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